Self Attention Mechanism
(Embedding Vector Dimension and Softmax Sensitivity)

$$w'_{ij} = x_i^T x_j$$

$[X_i]_{1\times d}$

d ↑

↑ $w'_{ij} = x_i^T x_j$

As Dimension of Embedding "d" Increases the Calculated weight value "w" increases

$[X_i]_{1\times d}$

d ↓

SoftMax $w_{ij} = \dfrac{\exp w'_{ij}}{\Sigma_j \exp w'_{ij}}$

High Weight Values kill the gradient, and slow down learning

d ↑ d ↑↑

w'_{ij}	SoftMax	For small "d"	w'_{ij}	SoftMax	For medium "d"	w'_{ij}	SoftMax	For large "d"
1	→	0.0321	5	→	0.0000	10	→	0.0000
2	→	0.0871	10	→	0.0000	20	→	0.0000
3	→	0.2369	15	→	0.0067	30	→	0.0000
4	→	0.6439	20	→	0.9933	40	→	1.0000